Moon Mission Plant and Flight Software: Mathematics

1 Conventions

SI units throughout: meters, seconds, kilograms, radians. Frame: origin at Earth's center, z along Earth's rotation axis, x toward the vernal equinox, axes fixed relative to the stars (Earth-centered equatorial). Source: `workloads/moon/common/`.

2 Vectors

For $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$ and scalar s:

$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z \tag{1}$$

$$\mathbf{a} \times \mathbf{b} = (a_y b_z - a_z b_y,\ a_z b_x - a_x b_z,\ a_x b_y - a_y b_x) \tag{2}$$

$$\|\mathbf{a}\| = \sqrt{\mathbf{a} \cdot \mathbf{a}} \tag{3}$$

The cross product follows the right-hand rule: $\hat{x} \times \hat{y} = \hat{z}$. Code: `vec3.hpp`. Tests: `test_vec3.cpp`.

3 Constants

Symbol	Value	Unit	Source
μ_E	$3.986004418 \times 10^{14}$	$\mathrm{m}^3/\mathrm{s}^2$	IERS Conventions 2010
μ_M	$4.902800066 \times 10^{12}$	$\mathrm{m}^3/\mathrm{s}^2$	JPL DE430
R_E	6 378 137	m	equatorial radius
R_M	1 737 400	m	mean radius
ω_E	7.292115×10^{-5}	rad/s	sidereal rotation
a_M	3.844×10^{8}	m	mean Earth–Moon distance
ε	23.439°		obliquity of the ecliptic
i_M	5.145°		lunar orbit inclination to the ecliptic
ϕ_L	28.573°		launch latitude, Kennedy LC-39A
g_0	9.80665	m/s^2	standard gravity

Code: `constants.hpp`.

4 Time

Mission time t is seconds since the epoch $\mathrm{JD}_0 = 2\,434\,743.5$ (1954-01-01 00:00 UTC; Bart Howard's original "Fly Me to the Moon" sheet music was published in 1954). Julian centuries from J2000:

$$T = \frac{\mathrm{JD}_0 + t/86400 - 2\,451\,545.0}{36\,525} \tag{4}$$

5 Moon ephemeris

Circular orbit of radius a_M. Mean elements, Meeus, *Astronomical Algorithms*, ch. 47, linear terms (higher-order terms change the result by about 0.0003° at the epoch):

$$L' = 218.3164477^\circ + 481\,267.88123421^\circ\, T \qquad \text{mean longitude} \tag{5}$$

$$\Omega = 125.0445479^\circ - 1934.1362891^\circ\, T \qquad \text{ascending node} \tag{6}$$

$$u = L' - \Omega \qquad \text{argument of latitude} \tag{7}$$

Position in the ecliptic frame:

$$\mathbf{r}_{\text{ecl}} = a_M \begin{pmatrix} \cos\Omega\cos u - \sin\Omega\sin u\cos i_M \\ \sin\Omega\cos u + \cos\Omega\sin u\cos i_M \\ \sin u\,\sin i_M \end{pmatrix} \tag{8}$$

Rotation into the Earth-equatorial frame about x by ε:

$$\mathbf{r}_M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\varepsilon & -\sin\varepsilon \\ 0 & \sin\varepsilon & \cos\varepsilon \end{pmatrix} \mathbf{r}_{\text{ecl}} \tag{9}$$

The orbit's tilt I to Earth's equator follows from

$$\cos I = \cos\varepsilon\cos i_M - \sin\varepsilon\sin i_M\cos\Omega \tag{10}$$

At the epoch, $I = 25.988^\circ$, within the real range 18.3°–28.6° over the 18.6-year nodal cycle.

Limitations. The real orbit is eccentric ($e \approx 0.055$), so the modeled position differs from the true one by up to about 6° in longitude and 20 000 km in distance. $u = L' - \Omega$ is the standard approximation. Code: `moon_ephemeris.hpp`. Tests: `test_moon_ephemeris.cpp`.